Storyline

Chapter 6: Circular Motion and Other Applications of Newton's Laws



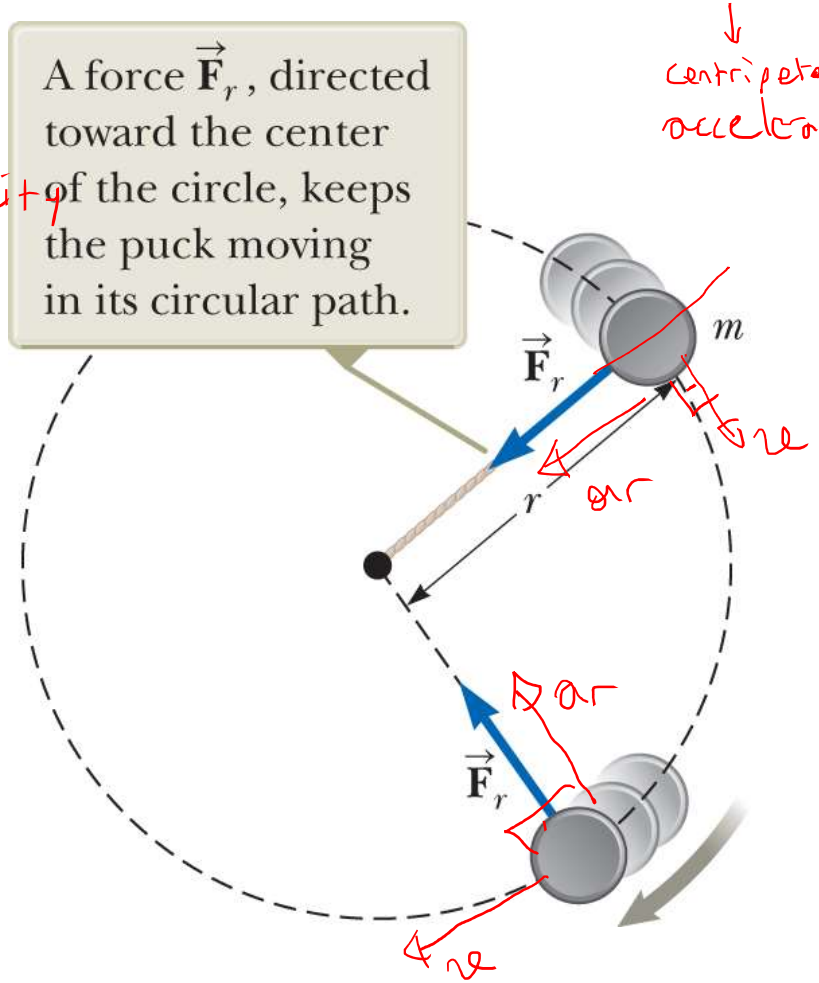
Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

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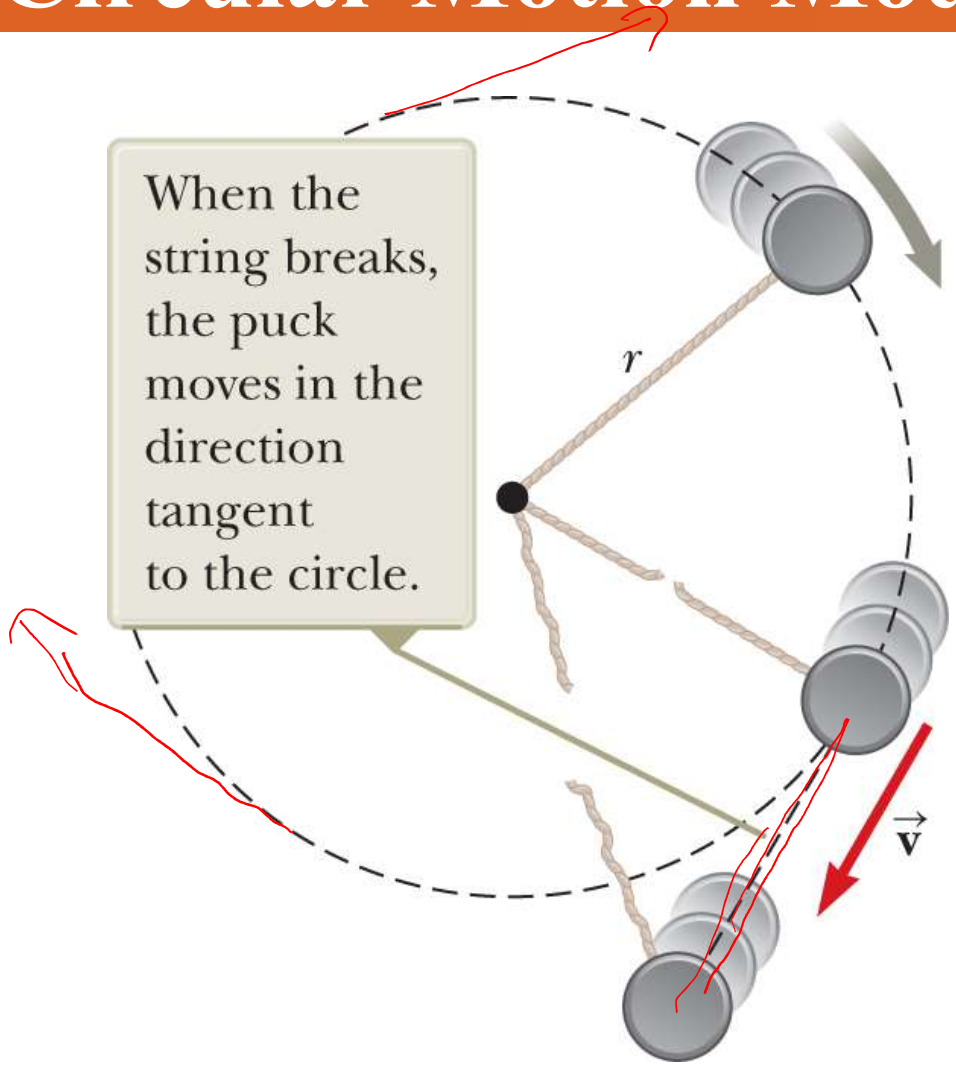
Extending the Particle in Uniform Circular Motion Model

$$a_c = \frac{v^2}{r}$$

$$\sum F = ma_c = m\frac{v^2}{r}$$



Extending the Particle in Uniform Circular Motion Model



Quick Quiz 6.1 Part I

You are riding on a Ferris wheel that is rotating with constant speed. The car in which you are riding always maintains its correct upward orientation; it does not invert.

What is the direction of the normal force on you from the seat when you are at the top of the wheel?

(a) upward

(b) downward

(c) impossible to determine



Quick Quiz 6.1 Part I

You are riding on a Ferris wheel that is rotating with constant speed. The car in which you are riding always maintains its correct upward orientation; it does not invert.

What is the direction of the normal force on you from the seat when you are at the top of the wheel?

(a) upward

(b) downward

(c) impossible to determine

Quick Quiz 6.1 Part II

You are riding on a Ferris wheel that is rotating with constant speed. The car in which you are riding always maintains its correct upward orientation; it does not invert.

What is the direction of the net force on you when you are at the top of the wheel?

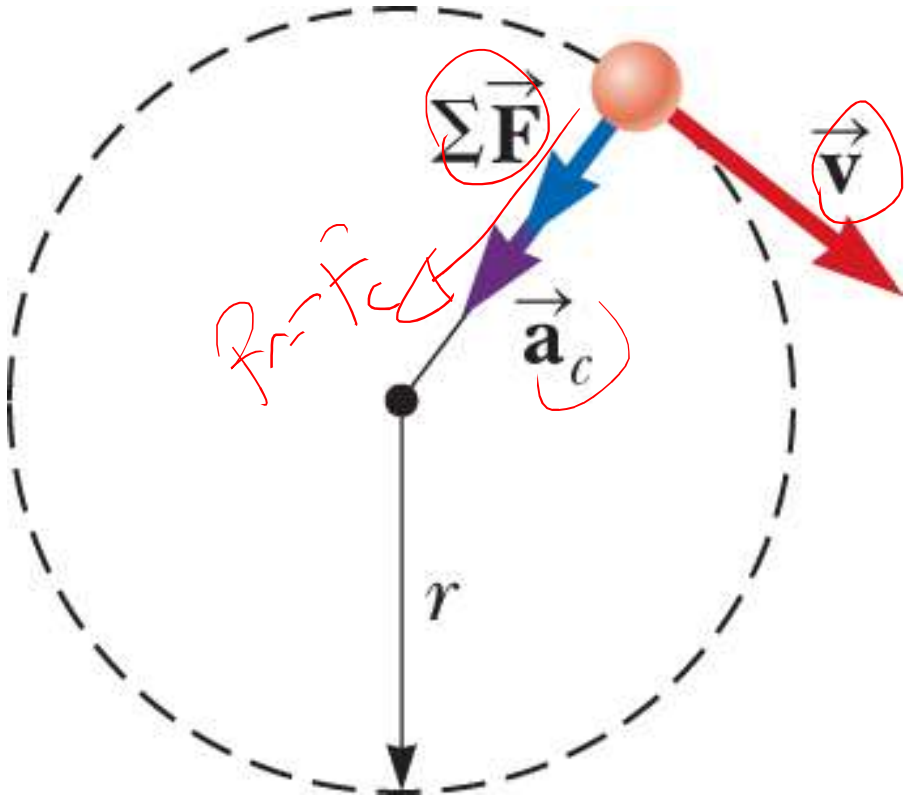
(a) upward

(b) downward

(c) impossible to determine

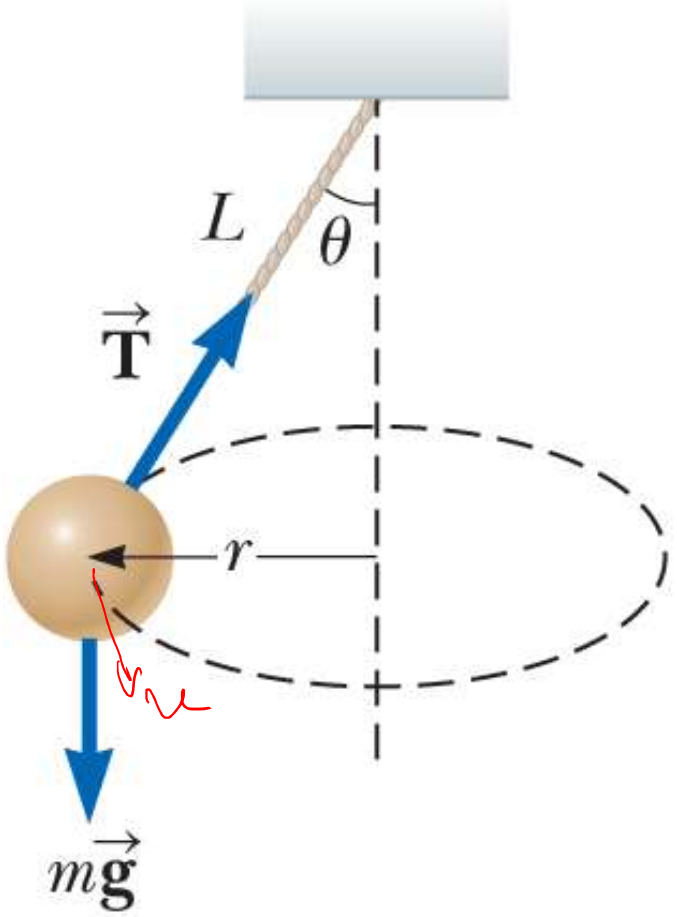
Analysis Model: Particle in Uniform Circular Motion (Extension)

$$\sum F = ma_c = m\frac{v^2}{r}$$



Example 6.1:
The Conical Pendulum

A small ball of mass m is suspended from a string of length L. The ball revolves with constant speed v in a horizontal circle of radius r as shown in the figure. (Because the string sweeps out the surface of a cone, the system is known as a *conical pendulum*.) Find an expression for v in terms of the length of the string and the angle it makes with the vertical in the figure.



Example 6.1: The Conical Pendulum

$$\sum F_y = T\cos\theta - mg = 0 \rightarrow T\cos\theta = mg \quad (1)$$

$$\sum F_x = T\sin\theta = ma_c = \frac{mv^2}{r} \quad (2)$$

Free body diagram.

$$\frac{(2)}{(1)} \qquad \tan\theta = \frac{v^2}{rg}$$

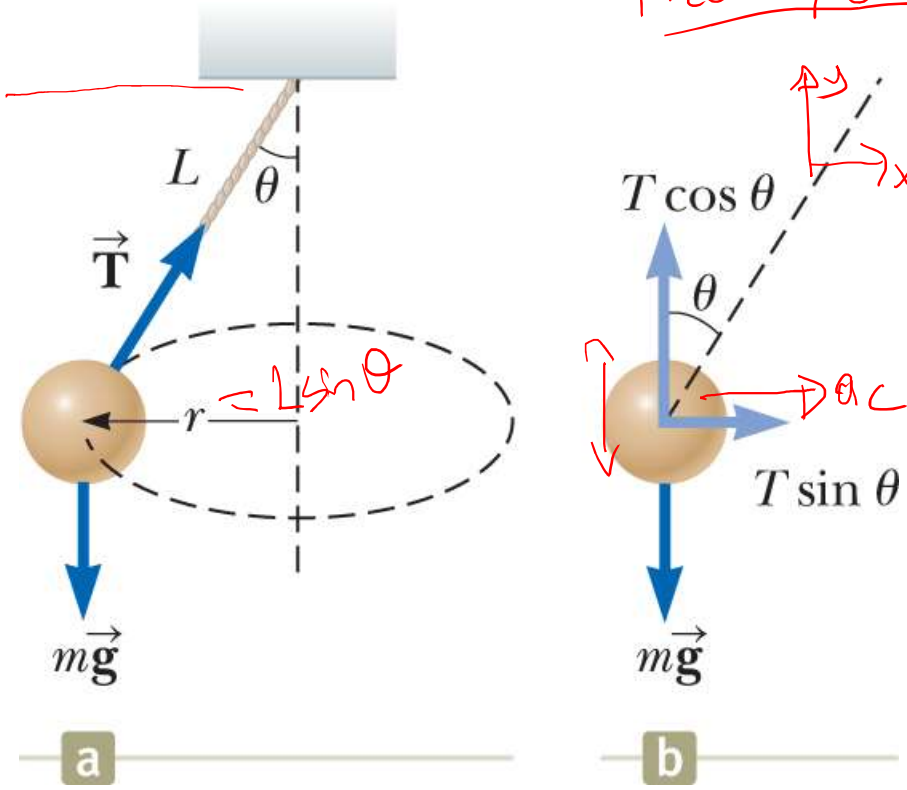
$$v = \sqrt{rg\tan\theta}$$

$r = L\sin\theta$

$$v = \boxed{\sqrt{Lg\sin\theta\tan\theta}}$$

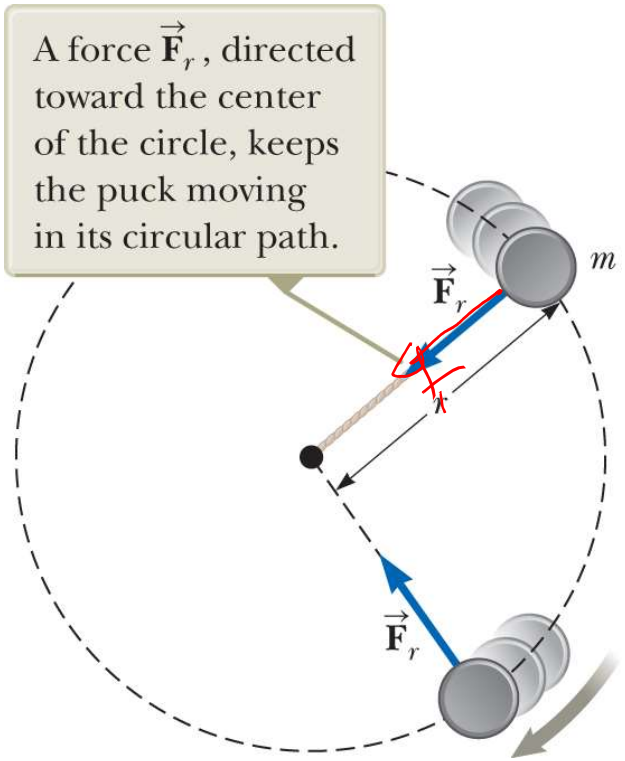
$\theta \rightarrow 90^\circ \Rightarrow \tan\theta \rightarrow \infty$

$v \rightarrow \infty$



Example 6.2:
How Fast Can It Spin?

A puck of mass 0.500 kg is attached to the end of a cord 1.50 m long. The puck moves in a horizontal circle as shown in the figure. If the cord can withstand a maximum tension of 50.0 N, what is the maximum speed at which the puck can move before the cord breaks? Assume the string remains horizontal during the motion.



Example 6.2:
How Fast Can It Spin?

If: $T = \frac{mv^2}{r}$

$$T = m\frac{v^2}{r}$$

$$v = \sqrt{\frac{Tr}{m}}$$

$$v_{\max} = \sqrt{\frac{T_{\max} r}{m}} = \sqrt{\frac{(50.0\ \text{N})(1.50\ \text{m})}{0.500\ \text{kg}}} = \boxed{12.2\ \text{m/s}}$$

Example 6.2:
How Fast Can It Spin?

Suppose the puck moves in a circle of larger radius at the same speed v. Is the cord more likely or less likely to break?

$v = \sqrt{\frac{Tr}{m}}$

less likely

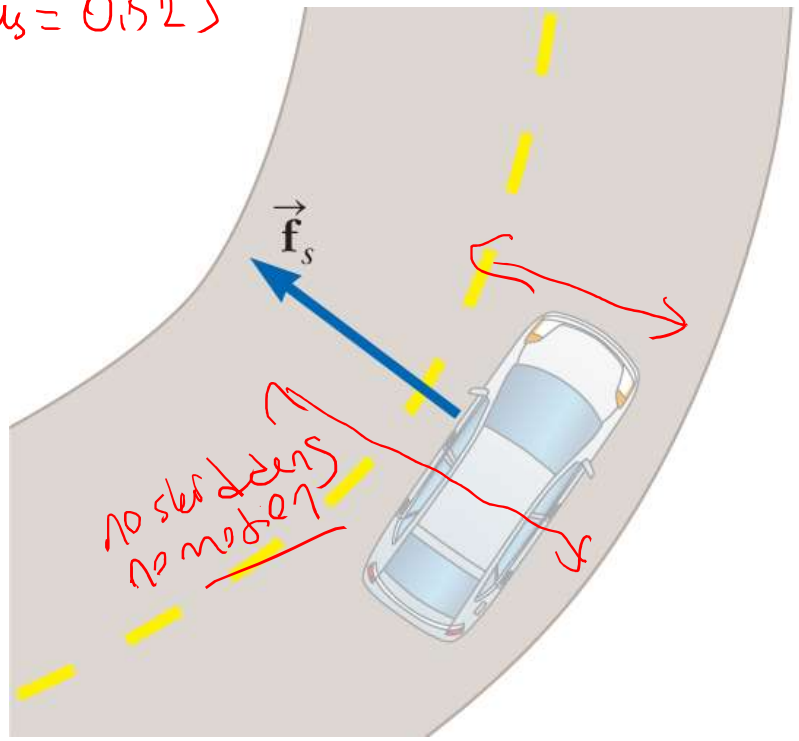


As r get larger the direction velocity vector will be smaller

$a_c = \frac{\Delta v}{r}$ $T = m a_c$

Example 6.3:
What Is the Maximum Speed of the Car?

A 1 500-kg car moving on a flat, horizontal road negotiates a curve as shown in the overhead view in the figure. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the maximum speed the car can have and still make the turn successfully.



Example 6.3:
What Is the Maximum Speed of the Car?

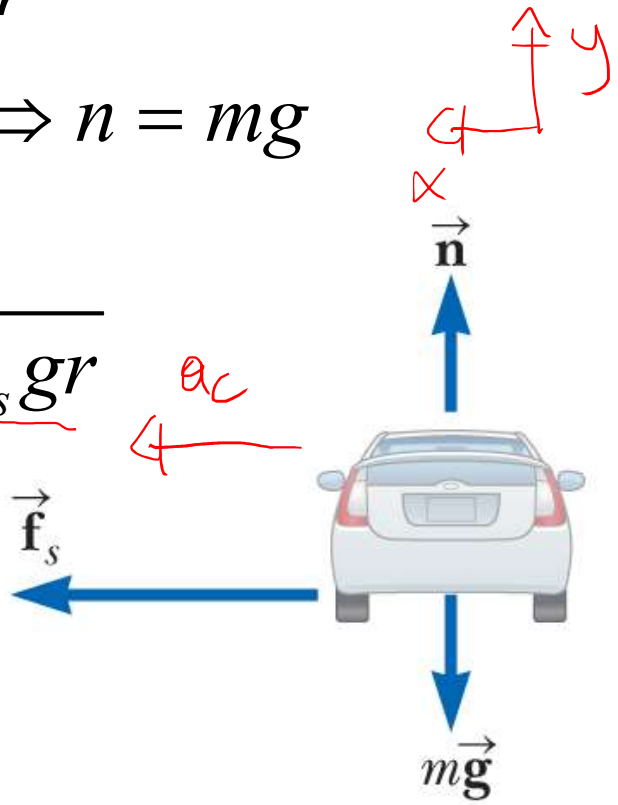
$\Sigma F_x = ma_c$

$$f_{s,\max} = \mu_s n = m\frac{v_{\max}^2}{r}$$

a_c

It is static because no slipping occurs between road and tires

$$\sum F_y = 0 \Rightarrow n - mg = 0 \Rightarrow n = mg$$



$$v_{\max} = \sqrt{\frac{\mu_s nr}{m}} = \sqrt{\frac{\mu_s mgr}{m}} = \sqrt{\mu_s gr}$$

If $v > v_{\max}$

$$v_{\max} = \sqrt{(0.523)(9.80\ \text{m/s}^2)(35.0\ \text{m})} = 13.4\ \text{m/s}$$

Example 6.3:
What Is the Maximum Speed of the Car?

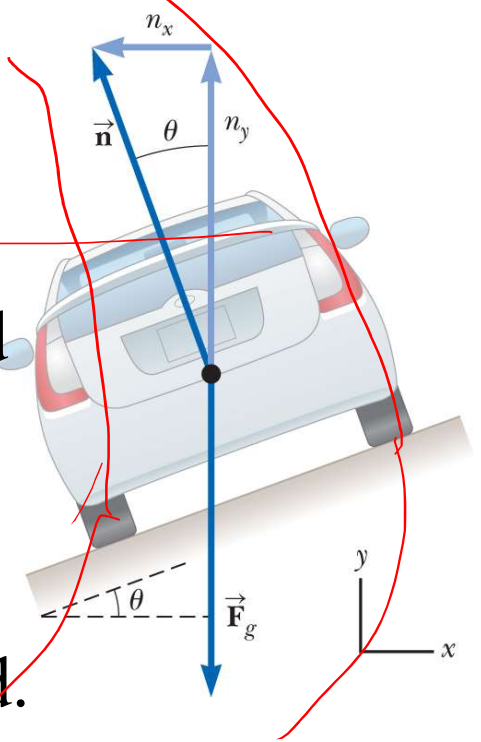
Suppose a car travels this curve on a wet day and begins to skid on the curve when its speed reaches only 8.00 m/s. What can we say about the coefficient of static friction in this case?

$$v_{\max} = \sqrt{\mu_s g r} \Rightarrow \mu_s = \frac{v_{\max}^{\ 2}}{gr}$$

$$\mu_s = \frac{(8.00\ \text{m/s})^2}{(9.80\ \text{m/s}^2)(35.0\ \text{m})} = 0.187$$

Example 6.4:
The Banked Roadway

You are a civil engineer who has been given the assignment to design a curved roadway in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve even when the road is covered with ice. Such a road is usually *banked*, which means that the roadway is tilted toward the inside of the curve. Suppose the designated speed for the road is to be 13.4 m/s (30.0 mi/h) and the radius of the curve is 35.0 m. You need to determine the angle at which the roadway on the curve should be banked.



Example 6.4:
The Banked Roadway

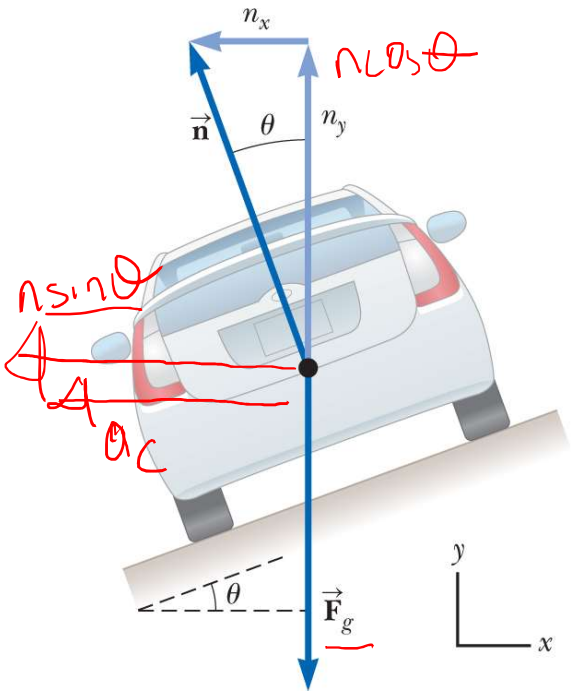
$$\sum F_r = n\sin\theta = \frac{mv^2}{r}$$

$$\sum F_y = n\cos\theta - mg = 0 \Rightarrow n\cos\theta = mg$$

$$\tan\theta = \frac{v^2}{rg}$$

$$\theta = \tan^{-1}\left[\frac{(13.4\ \text{m/s})^2}{(35.0\ \text{m})(9.80\ \text{m/s}^2)}\right]$$

$$= \boxed{27.6^\circ}$$

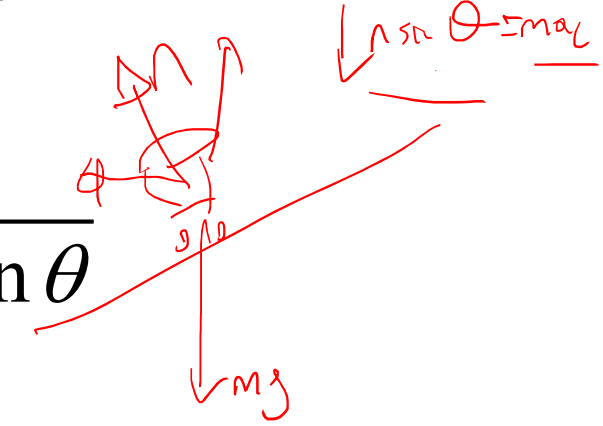


Example 6.4:
The Banked Roadway

Imagine that this same roadway were built on Mars in the future to connect different colony centers. Could it be traveled at the same speed?

No, *v* would be reduced.

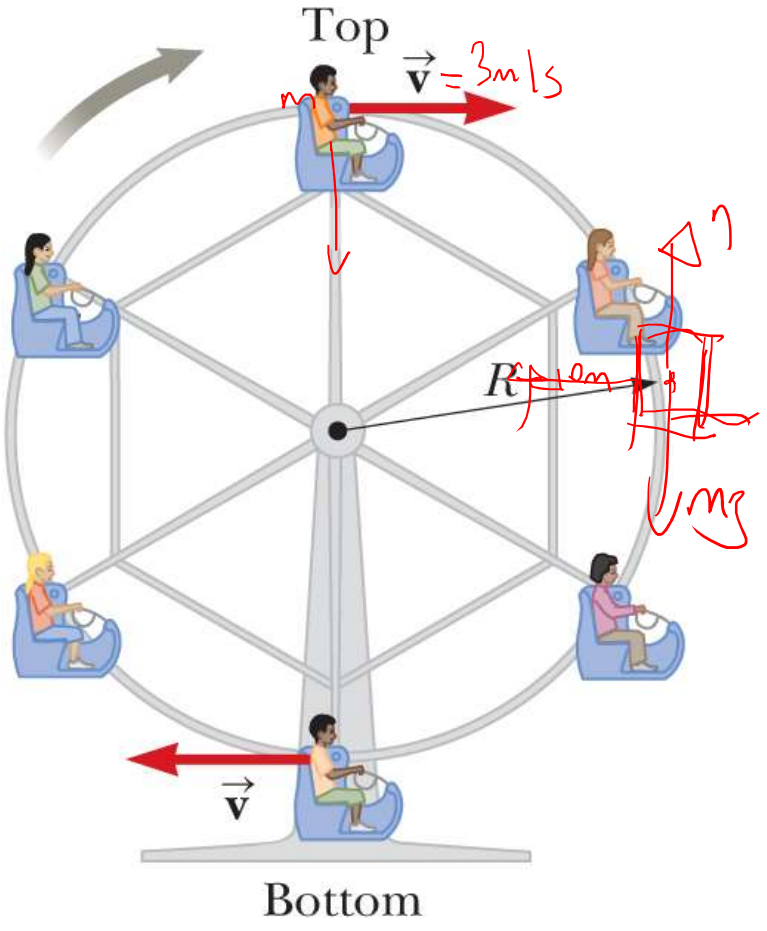
$$\tan\theta = \frac{v^2}{rg} \Rightarrow v = \sqrt{rg\tan\theta}$$



Example 6.5:
Riding the Ferris Wheel

A child of mass *m* rides on a Ferris wheel as shown in the figure. The child moves in a vertical circle of radius 10.0 m at a constant speed of 3.00 m/s.

(A) Determine the force exerted by the seat on the child at the bottom of the ride. Express your answer in terms of the weight of the child, *mg*.



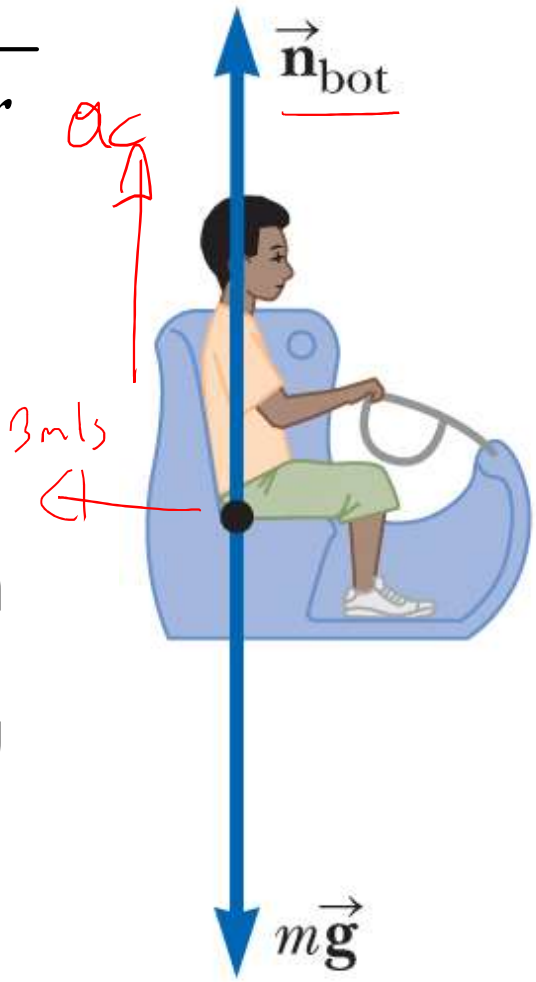
Example 6.5:
Riding the Ferris Wheel

$$\sum F = n_{\text{bot}} - mg = m\frac{v^2}{r}$$

$$n_{\text{bot}} = mg + m\frac{v^2}{r} = mg\left(1 + \frac{v^2}{rg}\right)$$

$$n_{\text{bot}} = mg\left(1 + \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right)$$

$$= \boxed{1.09mg}$$



Example 6.5:
Riding the Ferris Wheel

(B) Determine the force exerted by the seat on the child at the top of the ride.

$$\sum F = mg - n_{\text{top}} = m\frac{v^2}{r}$$

$$n_{top} = mg - m\frac{v^2}{r} = mg\left(1 - \frac{v^2}{rg}\right)$$

$$n_{top} = mg\left(1 - \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right)$$

$$= \boxed{0.908mg}$$



Example 6.5:
Riding the Ferris Wheel

Suppose a defect in the Ferris wheel mechanism causes the speed of the child to increase to 10.0 m/s. What does the child experience at the top of the ride in this case?

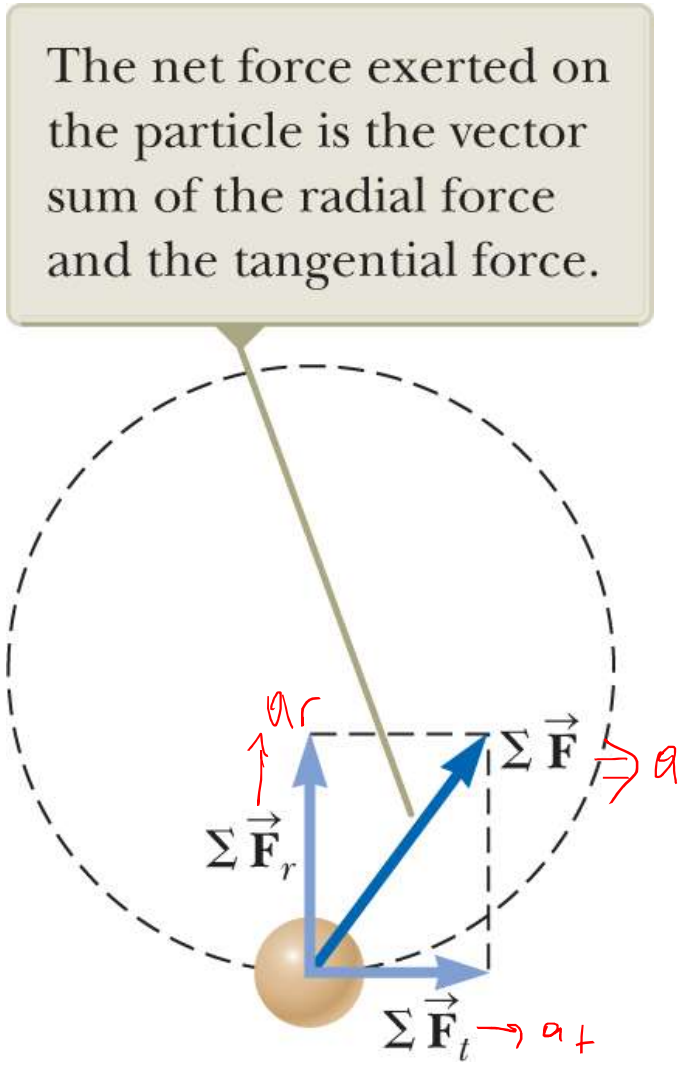
$$n_{top} = mg\left(1 - \frac{(10.0\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right) = -0.020mg$$

The child loses contact with his seat.

Nonuniform Circular Motion

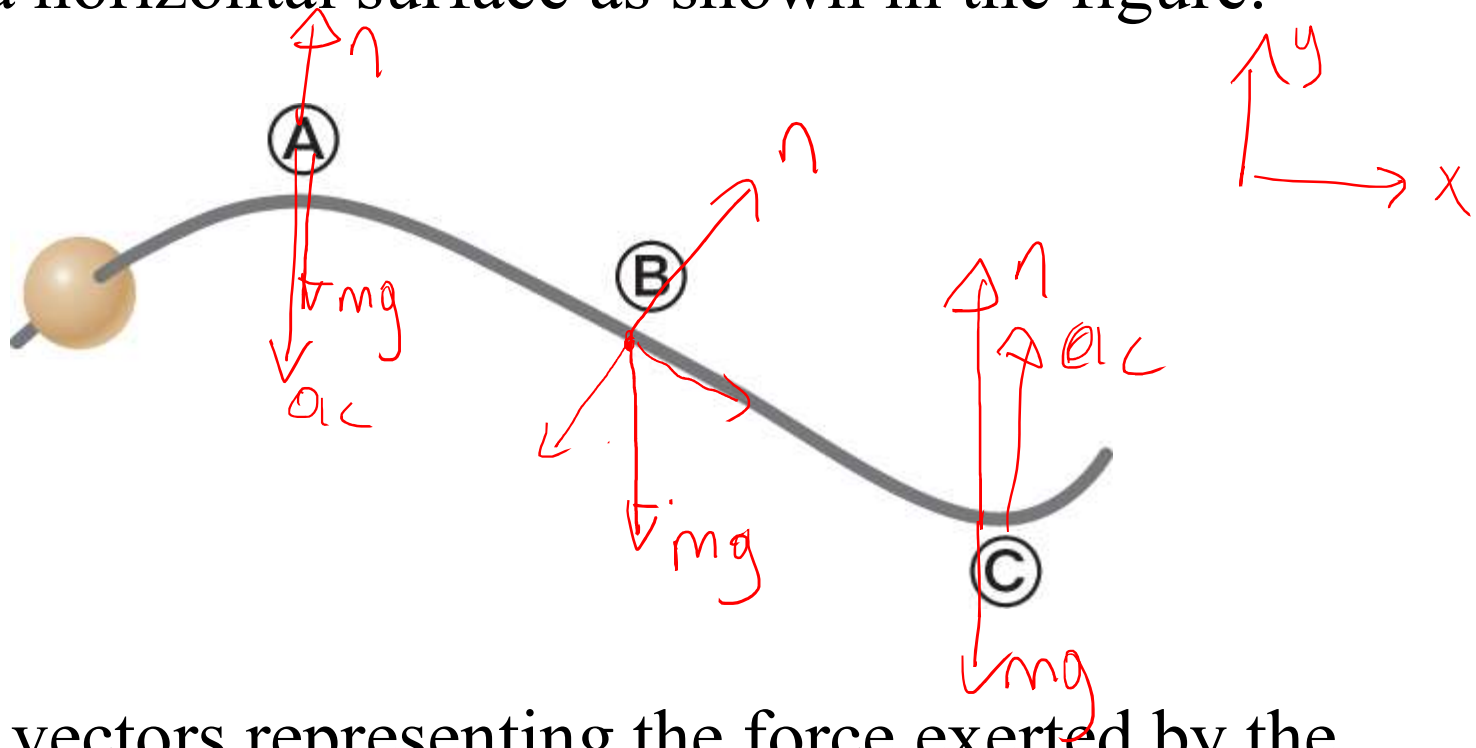
$$\vec{\mathbf{a}} = \vec{\mathbf{a}}_r + \vec{\mathbf{a}}_t$$

$$\sum \vec{\mathbf{F}} = \sum \vec{\mathbf{F}}_r + \sum \vec{\mathbf{F}}_t$$



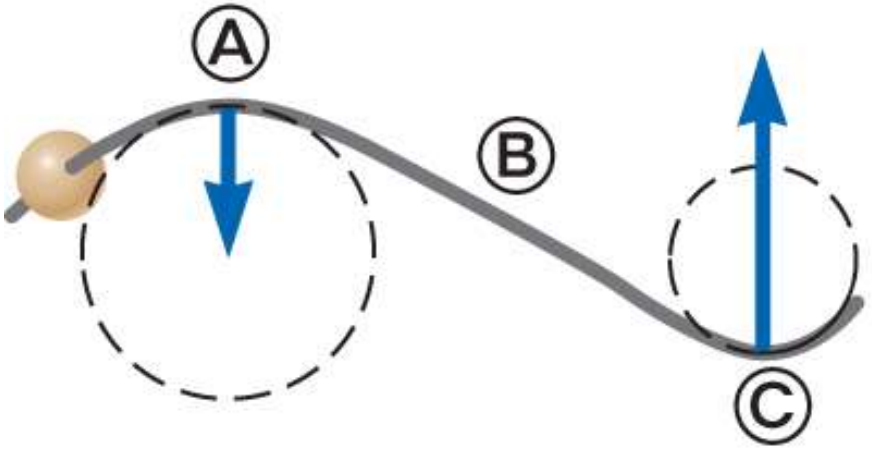
Quick Quiz 6.2 Part I

A bead slides at constant speed along a curved wire lying on a horizontal surface as shown in the figure.



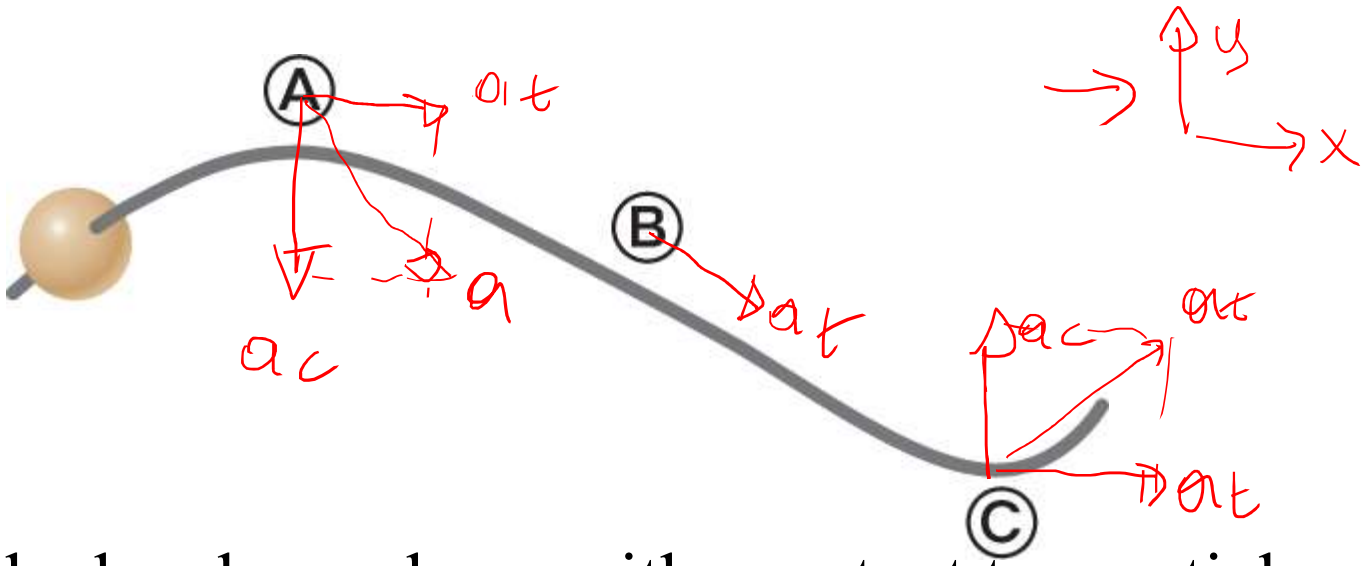
Draw the vectors representing the force exerted by the wire on the bead at points A, B, and C.

Quick Quiz 6.2 Part I



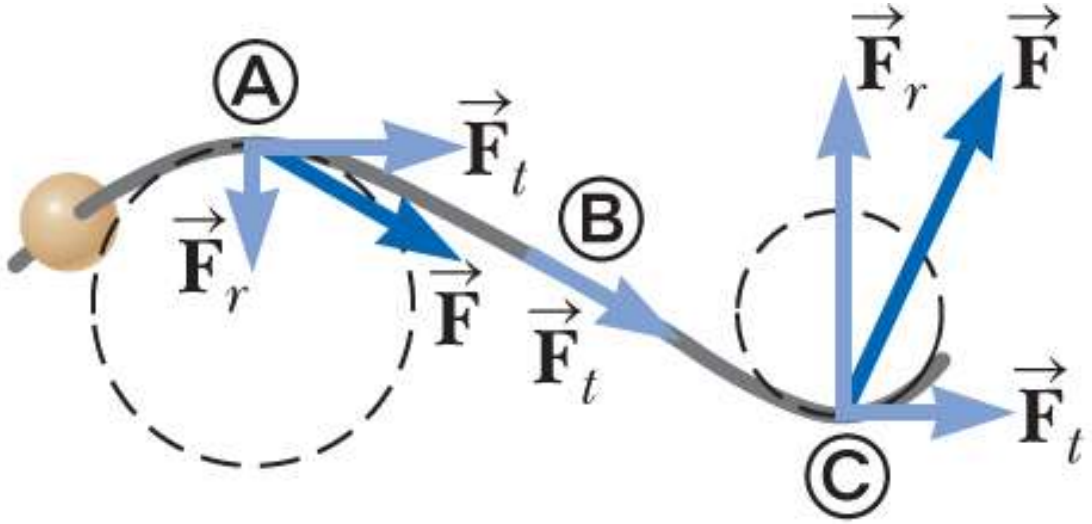
Quick Quiz 6.2 Part II

A bead slides at constant speed along a curved wire lying on a horizontal surface as shown in the figure.



Suppose the bead speeds up with constant tangential acceleration as it moves toward the right. Draw the vectors representing the force on the bead at points A, B, and C.

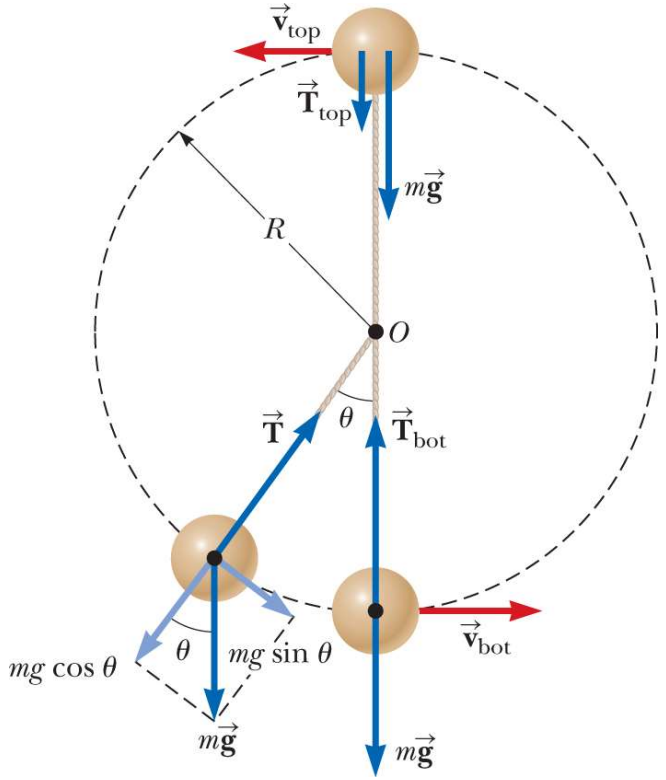
Quick Quiz 6.2 Part II



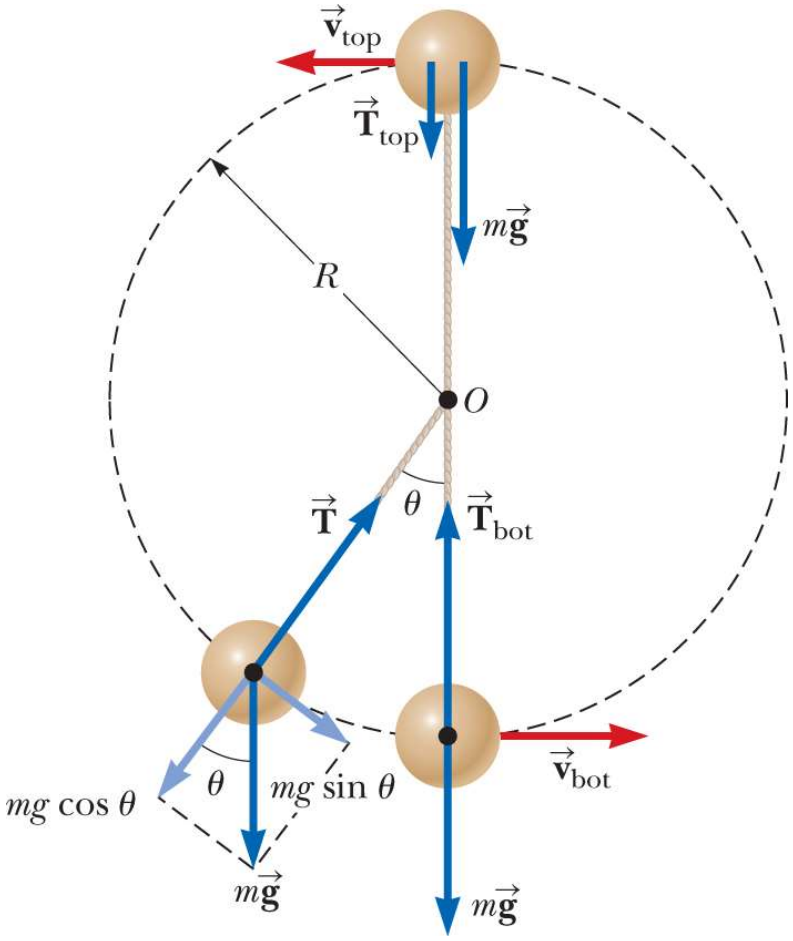
Example 6.6:
Keep Your Eye on the Ball

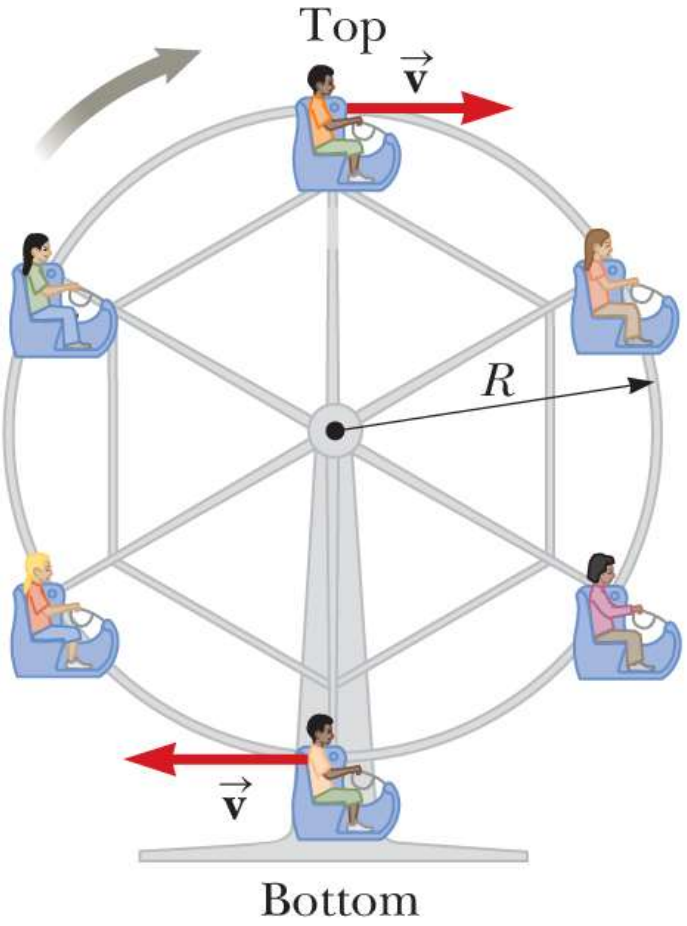
A small sphere of mass m is attached to the end of a cord of length R and set into motion in a *vertical* circle about a fixed point O as illustrated in the figure.

Determine the tangential acceleration of the sphere and the tension in the cord at any instant when the speed of the sphere is v and the cord makes an angle θ with the vertical.

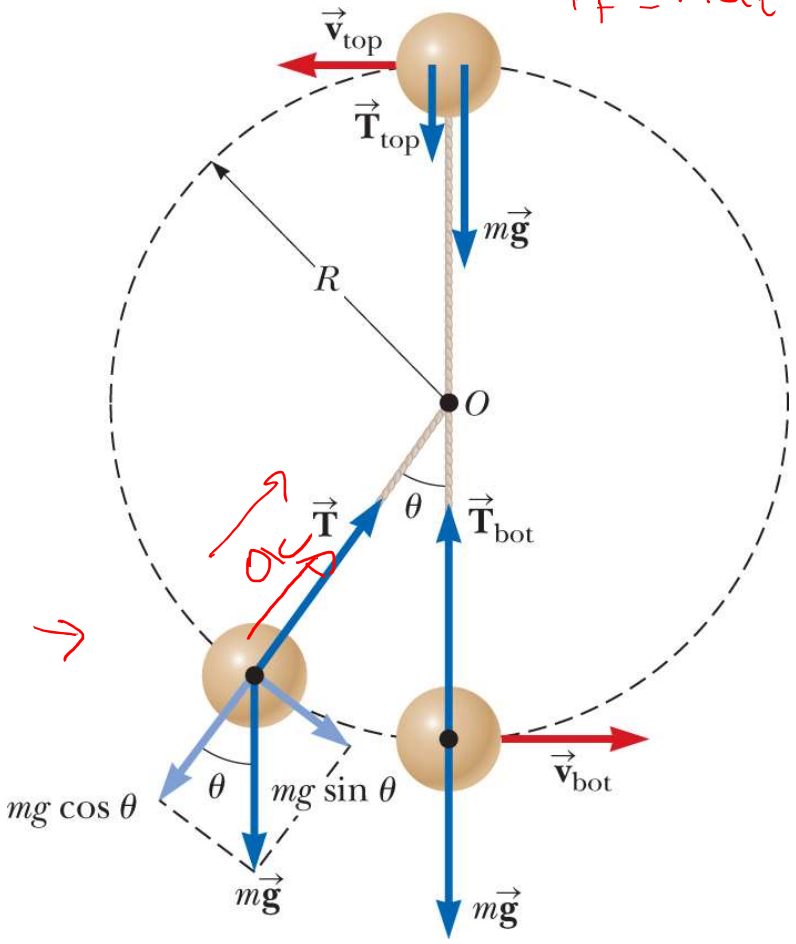


Example 6.6:
Keep Your Eye on the Ball





Example 6.6:
Keep Your Eye on the Ball



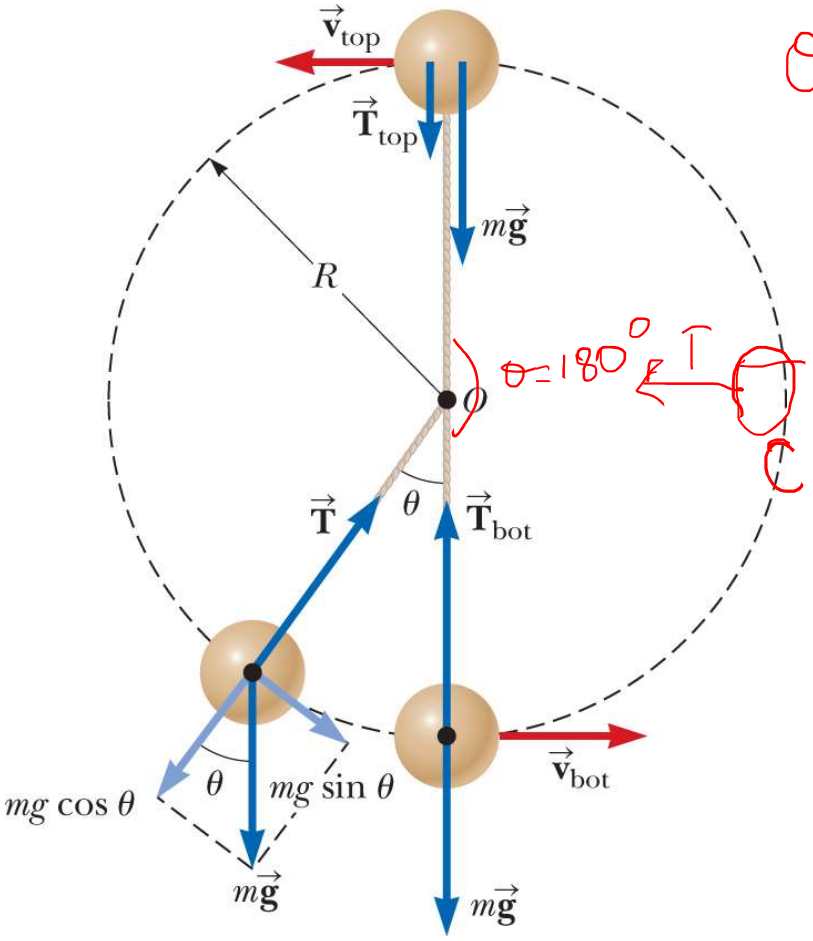
$$\sum F_t = mg\sin\theta = ma_t$$

$$a_t = \boxed{g\sin\theta}$$

$$\sum F_r = T - mg\cos\theta = \frac{mv^2}{R}$$

$$T = \boxed{mg\left(\frac{v^2}{Rg} + \cos\theta\right)}$$

Example 6.6:
Keep Your Eye on the Ball



$T = mg\left(\frac{v^2}{Rg} + \cos\theta\right)$

$\theta = 0$

$\theta = 180^\circ$

$$T_{\text{top}} = mg\left(\frac{v_{top}^{\ 2}}{Rg} - 1\right)$$

$$T_{\text{bot}} = mg\left(\frac{v_{\text{bot}}^{\ 2}}{Rg} + 1\right) \rightarrow \theta = 0$$

$T_c = \frac{mv^2}{R}$

Example 6.6:
Keep Your Eye on the Ball

(A) What speed would the sphere have if it passes over the top of the circle if the tension in the cord goes to zero instantaneously at this point?

$$T_{top} = mg\left(\frac{v_{top}^2}{Rg} + 1\right)$$

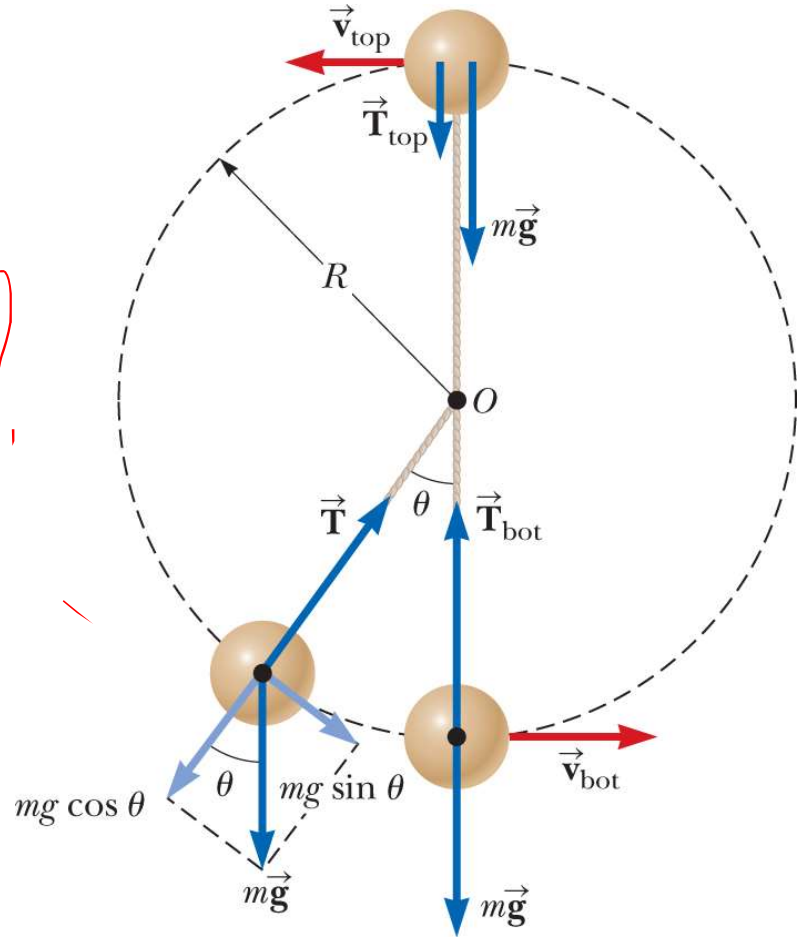
$$0 = mg\left(\frac{v_{top}^2}{Rg} - 1\right) \Rightarrow v_{top} = \sqrt{gR}$$

Example 6.6:
Keep Your Eye on the Ball

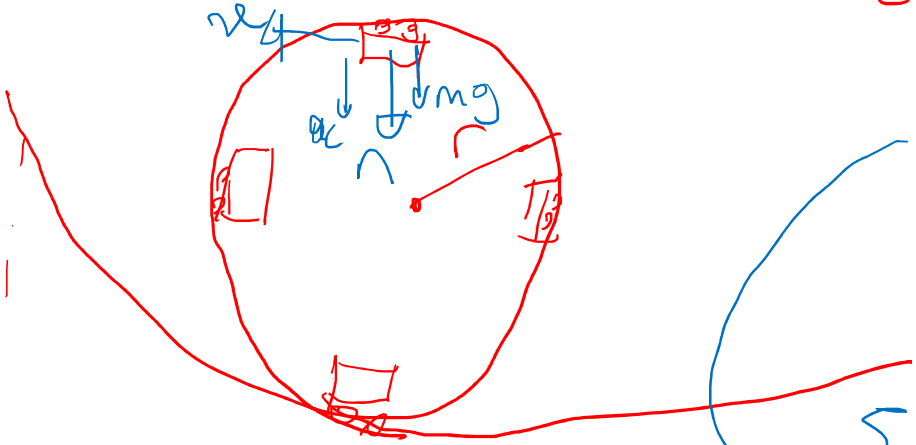
(B) What if the sphere is set in motion such that the speed at the top is less than this value? What happens?

If $v_{top} < v_{gt}$

The ball enters free fall for part of its motion near the top, and the string becomes slack during that time.



Circular Motion: Looping the Loop.



$r = 6.3\,m$. What is the minimum speed for roller-coaster car at the top of the loop if it is to stay on the track.

$\sum F = ma_c$

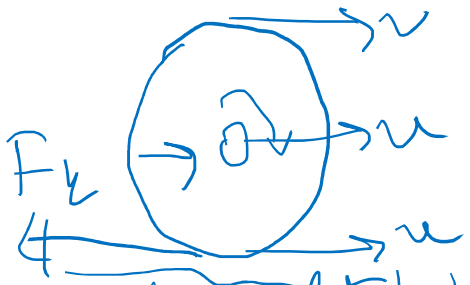
$n + mg = \frac{mv^2}{r}$ → ?

n ↓

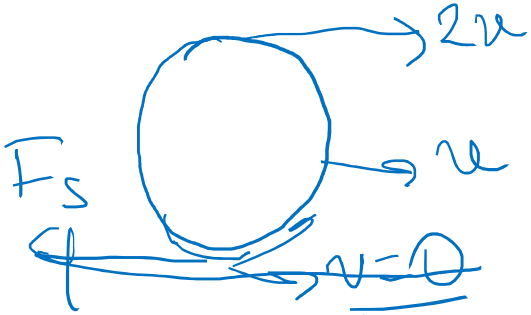
$0 + mg = \frac{mv^2}{r}$

$v_{min}^2 = rg = (6.3)10 \approx$ ~~8~~

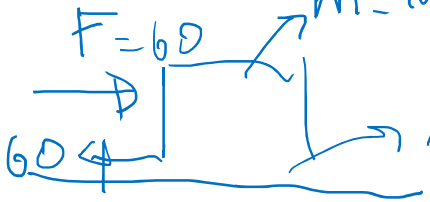
$v_{min} \approx$ ~~8~~ m/s



When a wheel skids, the force is kinetic friction



The bottom of rolling wheel is momentarily at rest, so the force is static friction



$F_{fmax} = \mu_s mg = 0,6 \cdot 10 \cdot 10 = 60N$

a) $F = 10N$, $Ff = 10N \rightarrow a = 0$

b) $F = 50N$, $Ff = 50N \rightarrow a = 0$

c) $F = 60N$, $Ff = ? \rightarrow 60N - Frict = 0 \rightarrow a = 0$

d) $F = 80N$, $Ff = ? \rightarrow \mu_k mg = 0,4 \cdot 10 \cdot 10 = 40N$